DAIICT

Tentative Syllabus for the New Course

ED112 (3-0-2-4)
ENGINEERING PHYSICS

Scope of the Course

Engineering Physics covers the basic mathematics and physics of oscillatory and wave phenomena.

After completing the course, students should be able to explain why oscillations appear in many near equilibrium systems, the various mathematical properties of those oscillations in various contexts, how oscillations and waves are related, and the basic mathematical description and properties of a wave. They will understand the interplay of oscillations in electronic and mechanical circuits and systems. Forced oscillation, role of friction and damping, and idea of resonance will be explained to them. They will understand the waves in transmission lines, and propagation of electromagnetic and sound waves in various media. Elementary quantum mechanical principles will also be introduced to them to appreciate the problem of harmonic oscillators.

This course also provides the students with a broad understanding of the physical principles of the oscillations, to help them develop critical thinking and quantitative reasoning skills, to empower them to think creatively and critically about scientific problems and experiments.

Possible Topics of Lectures and Experiments

Serial No.	TOPICS	Number of Lectures	Lab Experiments
1	Introduction to the course	1	

2	Conservation of Energy and Momenta, Potential and Kinetic Energy, Work, Force, etc. - Review	2	Familiarity with the lab. Oscilloscope, function generators, multimeters, breadboards
3	Simple Pendulum, Simple harmonic motion and Oscillation, Amplitude, frequency, phase and period of oscillations. Spring-mass system	1	Familiarity with SPICE software
4	Spring-Mass-Damper system, Damped Oscillation, Ordinary Differential Equation basics	2	Measurement of the Time period of an ideal pendulum
5	Review of R, L, C in terms of ODEs. Ohm's Law and I-V relations	1	

6	Solution of 1st order linear ODEs using integrating factors for various forcing functions, esp. step and sinusoidal inputs. Time constants. Transient and Steady-state responses	4	Measurement of RC and RL step and sinusoidal response
7	Laplace transform (one sided). Properties of LT. Inverse LT using table lookup. Solution of 1st order ODEs using LT.	3	Measurements with RLC circuits in both hardware and LTspice
8	Solution of 2nd order ODEs using LT. Over-, critical- and under-damped responses with step inputs.	2	Study of transmission lines using LTspice
9	Response of 2nd order ODEs for sinusoidal inputs. Damped and Forced Oscillations, Idea of Resonance and tuning	3	Standing wave measurements with a taught string

10	Oscillations, Waves. Amplitude, phase, frequency, wavelength, velocity of waves.	1	LED and Photodiode based experiment to prove inverse square law of intensity versus distance
11	Waves in transmission lines, Lossless transmission lines, telegraphist's equations, characteristic impedance and idea of reflection Solution of the lossless case – wave equation	3	
12	Reflection and standing waves, Standing wave ratio	1	
13	Waves of various types: Electromagnetic, Sound, Mechanical vibrations, Energy and Momentum	1	

14	Modern physics, Bohr theory of atoms, Wave-particle duality, Energy and momentum ideas with fundamental particles, De Broglie's idea	3	
15	Ideas of quantum mechanics, Uncertainty principle, Basic idea of probability	3	
16	Basic postulates of quantum mechanics, Schroedinger's equation, Idea of eigenvalue problems in ODEs, idea of the state of a particle	3	
17	Infinite potential well problem, idea of states and discrete energy levels	2	
18	Finite potential well problem, change in states and energy levels, tunneling ideas	2	

19	Harmonic oscillator problem, states, energy levels, series solution of ODEs, comparison with pendulum and RLC circuits	2	
20	Summary	1	
	Total	41	

Grading Policy

- Attendance and participation in Lectures: 5%. (Irrespective of everything else, the final grade would be an F if absent in more than 4 lectures in the whole semester.)
- Lab Experiments and reports: 20%. (Irrespective of everything else, the final grade would be an F if absent in more than 1 lab. in the whole semester.)
- In-sem. Exam. 1: 15%
- In-sem. Exam. 2: 20%
- End-sem. Exam: 30%
- Surprise Quizzes: 10%
- Irrespective of everything else, the final grade would be an F if any instance of copying, or cheating, or dishonesty is detected.

Reading Materials

- H.D. Young, R. A. Freedman, and A. L. Ford, University Physics with Modern Physics, 13th Edition, Addison-Wesley.
- S. O. Kasap, Principles of Electrical Engineering Materials and Devices, McGraw-Hill.

- Ebooks and documents distributed in the Google Classroom portal of the course.

NBA Program and Program-Specific Outcomes

P1 Eng. knowledge yes

P2 Problem analysis yes

P3 Design of solutions yes

P4 Investigation of problems yes

P5 Modern tool usage yes

P8 Ethics yes

P9 Ind. and team work yes

P12 Life-long learning yes

PSO 1 yes

PSO 2 yes

PSO 3 yes